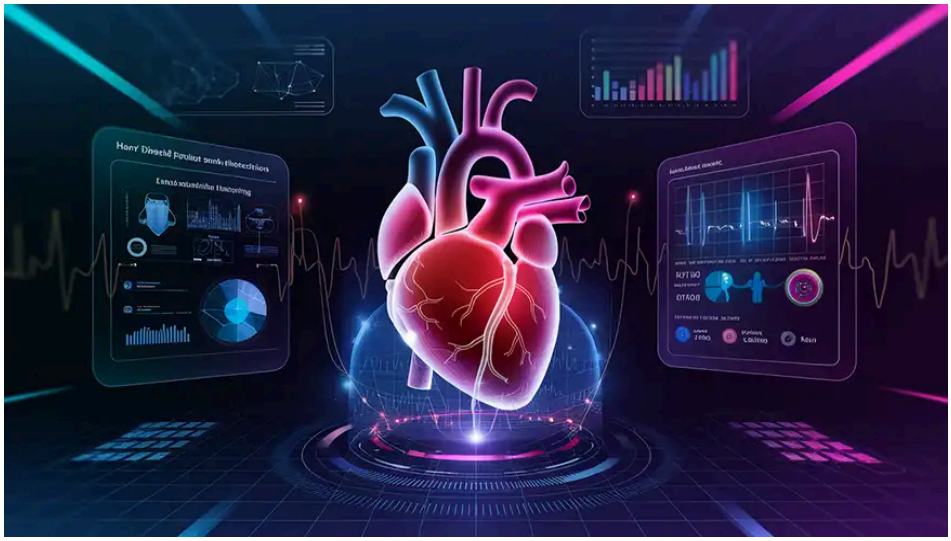
HEART DISEASE PREDICTION USING MACHINE LEARNING



⚫ INTRODUCTION

- Heart disease is one of the leading causes of death worldwide
- Early prediction can help in timely diagnosis and treatment
- Machine Learning (ML) helps analyze patient data efficiently

Project Objectives: Analyze patient data to identify key health factors and build an accurate ML model for heart disease prediction.

⚫ DATA COLLECTION

- Total Features: 14
- Dataset: Kaggle Heart Disease
- Total Records: 1025 patients

Features: age, sex, chest pain `cp`, cholesterol `chol`, and maximum heart rate `thalach`.

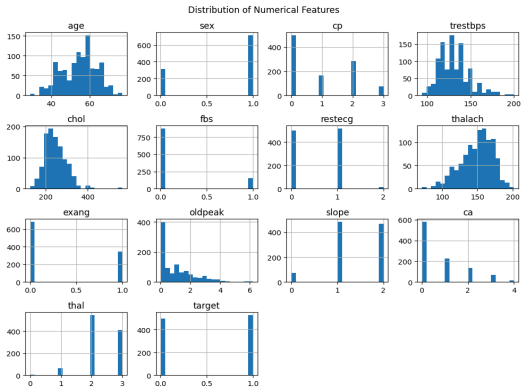
Target: Binary variable for heart disease (1=present, 0=absent) with 517(+) and 492(-) cases.

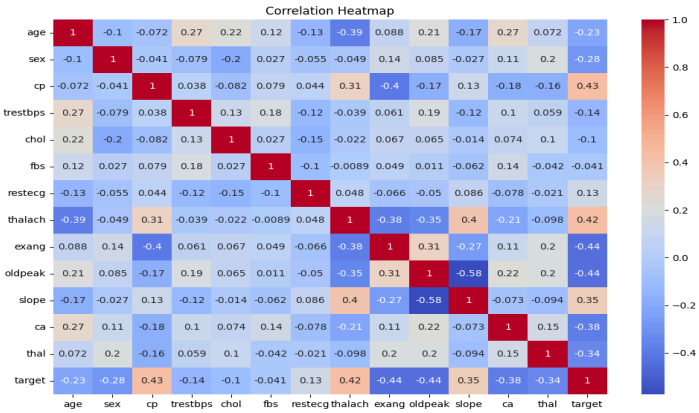
⚫ DATA PREPROCESSING

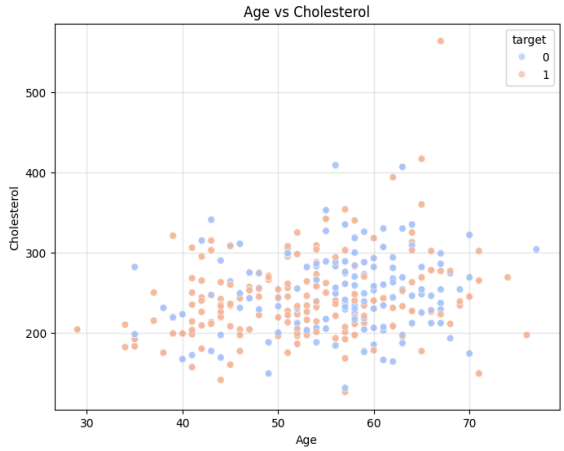
- checked for missing values and outliers • No missing data found, outliers retained due to medical importance.
- Dataset split into 80% training and 20% testing sets • Features scaled using StandardScaler.

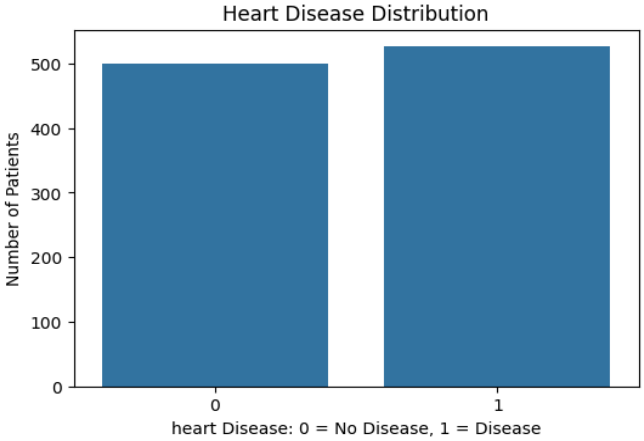
⚫ EXPLORATORY DATA ANALYSIS (EDA)

- Histograms → Most patients aged **40–65**
- Boxplots → Outliers in cholesterol & BP
- Heatmap → Strong feature correlations
- Scatter Plot → Weak relation (age vs cholesterol)
- Countplot → Balanced data









⚫ FEATURE ENGINEERING

- Created new feature: **age_chol_ratio**
- Combines age + cholesterol data
- Helps capture deeper patterns

After Processing

- X → Input features , y → Target variable

⚫ MODEL BUILDING

Three models were applied:

1. Logistic Regression:

- Simple baseline model.
- Fast and efficient

2. Decision Tree:

- Handles non-linear data
- Easy to interpret

3. Random Forest:

- Ensemble technique
- Combines multiple trees
- Reduces overfitting.

⚫ MODEL EVALUATION

Evaluation Metrics:

- Accuracy Score
- Confusion Matrix
- Classification Report

Results

- Random Forest → Highest Accuracy
- Better generalization
- Reduced overfitting

⚫ CONCLUSION

- ML successfully used for heart disease prediction
- Data preprocessing improved model performance
- Feature engineering enhanced accuracy
- **Random Forest performed best**
- 📌 Can be used in real-world healthcare for early detection

⚪ FUTURE IMPROVEMENTS

- Apply advanced models (SVM, XGBoost)
- Perform hyperparameter tuning
- Use larger datasets
- Apply cross-validation

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